

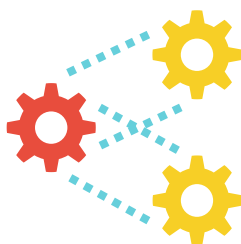
Implementing JWT

Using the Flask Framework

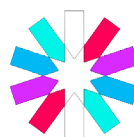
JSON web token, abbreviated as JWT, uses JSON objects as tokens. This article is an introduction to JWT and how to implement it using the Flask framework. It assumes you have a basic knowledge of Flask.



Flask



1 3 4 5 9 6 7 2



JWT

A token, as defined in Wikipedia, is an object that represents the right to perform some action. In client server architecture, the server generates a signed token and stores it in the client browser. Just like a cookie, the client has to pass this token with each request it makes to the server. The server then verifies the signature of the token and grants access to client requests on that basis. A token contains all the data within itself so that the server doesn't need to record any of the client data. It simply authorises client requests by relying on the validity of the token.

What is JWT?

JSON web token, abbreviated as JWT, uses JSON objects as tokens. JWT comprises three parts separated by dots (.). These are listed below.

- **Header:** This consists of two parts — the signing algorithm used (like HMAC or RSA) and the type of token, which by default is JWT.
- **Payload:** This part contains the claims and any additional data that needs to be encoded. Header and payload are Base64Url encoded.
- **Signature:** A signature is created by supplying the encoded header and payload along with a secret key to the algorithm specified in the header. A signature verifies that the message hasn't changed along the way.

To read more about JWT, you can refer to the official documentation at <https://jwt.io/introduction>.

Note that the token is only encoded and not encrypted. Thus, storing passwords inside a token can cause a security breach.

Implementing JWT with various encoding algorithms using Flask

HS256: This is a very popular algorithm since it is easy to implement. The secret key used here is a long sequence of characters such that it cannot be easily identified.

Step 1: First, install the `pyjwt` package using `pip`:

```
pip install pyjwt
```



Note: Make sure to activate the virtual environment, if any, in your project before installing the package.

Step 2: Create a file as `app.py` and paste the code given below:

```
from flask import Flask, session, url_for, redirect
import jwt
import datetime
import time

app = Flask(__name__)
app.secret_key = 'super secret key'
base_url = "http://127.0.0.1:5000"
secret_key = "mysecret"

@app.route('/')
@app.route('/<result>')
def home(result=None):
```